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实验名称：电位差计及其应用

实验一：自组电位差计测电动势. 预习-4

1. 原始数据记录与处理.

$t = 24.0^\circ C \quad E_{20} = 1.01862\ V$

$E_N = E_{20} - 3.99\times10^{-5}(t-20^\circ C) - 0.94\times10^{-6}(t-20^\circ)^2 - 9.5\times10^{-9}(t-20^\circ C)^3$

$= 1.01862 - 3.99\times10^{-5}\times(24-20) - 0.94\times10^{-6}(24-20)^2 - 9.5\times10^{-9}(24-20)^3$

$= 1.01844\ V.$

(电流标准化) $I_0 \equiv 1\ mA \quad R_1 = \frac{E_N}{I_0} = 1018.4\ \Omega$

类别	R_1/Ω	R_2/Ω	R_1'/Ω	R_2'/Ω
示值 $R_i(R_i')$	1018.4	1522.4	1433.0	1507.8
仪器误差 $\Delta R_i(\Delta R_i')$	1.105	1.595	1.50	1.60
灵敏度测量 (n=10 div)	/	/	1441.8	1499.0

$\Delta_{仪}(R_1) = 0.1\%\times1000 + 0.1\%\times0 + 0.2\%\times10 + 0.5\%\times8 + 5\%\times0.4 + 0.025 = 1.105\ \Omega$

$\Delta_{仪}(R_2) = 0.1\%\times1000 + 0.1\%\times500 + 0.2\%\times20 + 0.5\%\times2 + 5\%\times0.4 + 0.025 = 1.595\ \Omega$

$\Delta_{仪}(R_1') = 0.1\%\times1000 + 0.1\%\times400 + 0.2\%\times30 + 0.5\%\times3 + 5\%\times0 + 0.025 = 1.50\ \Omega$

$\Delta_{仪}(R_2') = 0.1\%\times1000 + 0.1\%\times500 + 0.2\%\times0 + 0.5\%\times7 + 5\%\times0.8 + 0.025 = 1.60\ \Omega$

$\therefore u(R_1) = \frac{\Delta_{仪}(R_1)}{\sqrt{3}} = 0.6380\ \Omega \quad u(R_2) = \frac{\Delta_{仪}(R_2)}{\sqrt{3}} = 1.1518\ \Omega$

$u(R_1') = \frac{\Delta_{仪}(R_1')}{\sqrt{3}} = 0.8660\ \Omega \quad u(R_2') = \frac{\Delta_{仪}(R_2')}{\sqrt{3}} = 0.9238\ \Omega$

$E_{(x)} = \frac{E_N}{R_1}\cdot R_1' = 1.433056\ V$

2. 计算仪器灵敏度与误差

(1) 灵敏度 $S = \frac{10}{I_0|R_1'-R_1''|} = \frac{10}{10^{-3}\times|1433-1441.8|} = 1136.4\ div/V$

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(2) 灵敏度误差 $\Delta_{灵}(E_x) = \frac{0.2}{S} = 1.76\times10^{-4}$ (V).

$U_{灵}(E_x) = \frac{\Delta_{灵}}{\sqrt{3}} = 1.016\times10^{-4}$ V

3. 计算不确定度.

$$\frac{U(E_x)}{E_x} = \sqrt{\left(\frac{1}{R_1}-\frac{1}{R_1+R_2}\right)^2 U^2(R_1) + \left(\frac{U(R_2)}{R_1+R_2}\right)^2 + \left(\frac{1}{R_1'}-\frac{1}{R_1'+R_2'}\right)^2 U^2(R_1') + \left(\frac{U(R_2')}{R_1'+R_2'}\right)^2}$$

$$= \frac{1}{R_1+R_2}\sqrt{\left(\frac{R_2}{R_1}\right)^2 U^2(R_1) + U^2(R_2) + \left(\frac{R_2'}{R_1'}\right)^2 U^2(R_1') + U^2(R_2')}$$

$$= \frac{1}{1018.4+1922.4}\sqrt{\left(\frac{1922.4}{1018.4}\times0.6380\right)^2 + (1.1518)^2 + \left(\frac{1922}{1441.8}\times0.866\right)^2 + (0.8238)^2}$$

$$= 7.166\times10^{-4}$$

$\therefore U(E_x) = \frac{U(E_x)}{E_x}\cdot E_x = 7.166\times10^{-4}\times1.433056 = 1.027\times10^{-3}$ V.

没有合入u灵 -5

4. 最终结果表达

$E_x \pm U(E_x) = (1.433 \pm 0.001)$ V.

5. 计算相对误差.

UJ25型箱式电位差计测量电池电动势为 1.461476 V

相对误差 $\gamma = \frac{|1.461476-1.433056|}{1.461476}\times100\% = 1.9\%$.

实验二: UJ25箱式电位差计测干电池电动势.

测量结果 $E_x = 1.461476$ V

计算不确定度: $\Delta_{仪} = a\%\left(V_x + \frac{U_0}{10}\right) = 0.01\%\times\left(1.461476 + \frac{0.1}{10}\right) = 1.47\times10^{-4}$ V

$U_{仪} = \frac{\Delta_{仪}}{\sqrt{3}} = 8.4956\times10^{-5}$ V

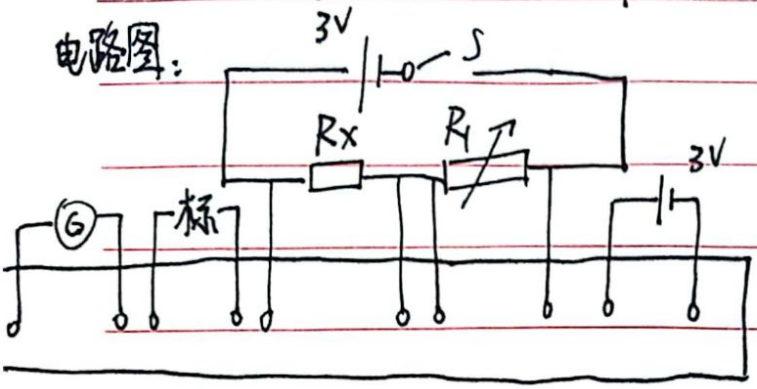
最终结果表达: $E_x \pm U_{Ex} = (1.46148 \pm 0.00008)$ V

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实验三：UJ31箱式电位差计测固定电阻.

电路图：



1. 原始数据记录与处理.

调节 $R_1 = 180\,\Omega$

U_x / V	U_N / V
1.48748	1.45053

$$R_x = \frac{U_x}{U_N} \times R_1 = 184.8\,\Omega$$

2. 计算误差和不确定度.

$$\Delta R_1 = 0.1\% \times 100 + 0.2\% \times 80 + 0.025 = 0.285\,\Omega$$

$$\Delta_{仪}(U_N) = a\%\left(U_N + \frac{U_N}{10}\right) = 0.01\% \times \left(1.45053 + \frac{0.1}{10}\right) = 1.46053 \times 10^{-4}\,V$$

$$\Delta_{仪}(U_x) = a\%\left(U_x + \frac{U_x}{10}\right) = 0.01\% \times \left(1.48748 + \frac{0.1}{10}\right) = 1.48748 \times 10^{-4}\,V$$

$$u(R_1) = \frac{\Delta R_1}{\sqrt{3}} = 0.165\,\Omega$$

$$u(U_N) = \frac{\Delta_{仪} U_N}{\sqrt{3}} = 0.844 \times 10^{-4}\,V$$

$$u(U_x) = \frac{\Delta_{仪} U_x}{\sqrt{3}} = 0.866 \times 10^{-4}\,V$$

$$\frac{u(R_x)}{R_x} = \sqrt{\left(\frac{u(R_1)}{R_1}\right)^2 + \left(\frac{u(U_N)}{U_N}\right)^2 + \left(\frac{u(U_x)}{U_x}\right)^2} = 1.203 \times 10^{-4}$$

$$\therefore u(R_x) = \frac{u(R_x)}{R_x} \cdot R_x = 1.203 \times 10^{-4} \times 184.8 = 0.170\,\Omega$$

3. 最终结果表达

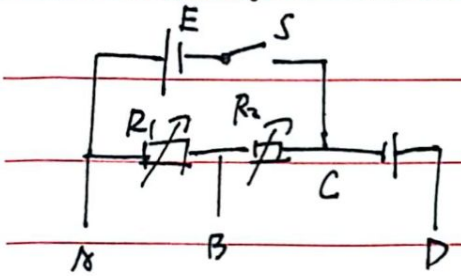
$$R_x \pm u(R_x) = (184.8 \pm 0.2)\,\Omega$$

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课后思考题

(1) 如何使用UJ25型箱式电阻差计测量约为4.5V的电源电动势.



取 $E=4.5V$ $R_1=1000\Omega$ $R_2=2000\Omega$

E_x 为待测电阻

如图接线. 可解决UJ25量程过小的问题

此时 U_{AB}. U_{BD} 均约1.5V可测.

$$\Rightarrow E_x = U_{BD} + U_{AB}\cdot\frac{R_2}{R_1}.$$

(2). 自组电位差在电学量及非电学量精确测量中的拓展及应用.

① 电学量测量.

电压测量: 自组电位差计高精度应用于电压测量

电阻测量: 通过与已知电阻和电源串联测量电阻

仪器校准: 可利用其高精度校准电学仪器.

② 非电学量测量.

温度测量、与热电偶等温感元件结合测量温度.

应力/应变测量: 与应变片相结合测量材料的应力和应变.

位移测量: 与线性电位计等位移传感器结合测量位移.

光强度测量: 与光电二极管或光敏电阻结合测量光强度.

材料表征: 对于一些半导体材料. 可用电位差计测量禁带宽度

载流子浓度等参数.

自动控制系统: 反馈和测量电路状态. 通过监测系统的电路

变化. 并将其与预定值比较. 实现自动调节和控制.

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总分91

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$t = 24.0^\circ C$ $E_N = 1.01862\,V$

	R_1/Ω	R_2/Ω	R_1'/Ω	R_2'/Ω
R_i (R_i')	1018.4	~~1995~~ 1022.4	1433.0	1507.8
ΔR_i ($\Delta R_i'$)				
$n = 10$ div	/	/	1041.8	1044.0

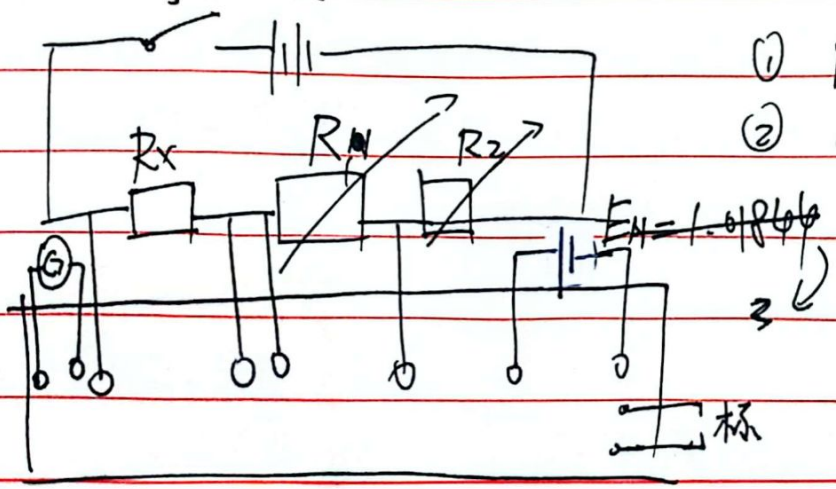
$E_N = E_{20} - 3.99\times10^{-5}\times(24-20) - 0.94\times10^{-6}\times(24-20)^2 + 9\times10^{-9}(24-20)^3 = 1.01844\,V$

$E_x = \frac{E_N}{R_1}\times R_1' = \frac{1.01844}{1018.4}\times 1433 = 1.433056\,V$

$R_1 = 1018.4$ $R_2 = 1022.4$ $R_1 + R_2 = 2040.8$

(2) $E_x = 1.46147\,6\,V$

(3)



① $R_x + R_1$

② $R_1 + R_2$ $\Rightarrow R_x = R_2$

184.8

2022.9.23

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